Experiment-7

Aim : To study and implement **Transaction Control Language (TCL)** commands such as:-

- ✓ **COMMIT** and **ROLLBACK** using a *Teacher* table.
- ✓ **COMMIT** → Saves all changes permanently in the database.
- ✓ **ROLLBACK** → Undoes changes made in the current transaction.
- ✓ These commands help maintain **data consistency** and **error recovery**.

Procedure & Queries

Step 1: Create database Lab_Assignment_7 and use database Lab_Assignment_7

```
CREATE DATABASE Lab_Assignment_7;
▷Run
USE Lab_Assignment_7;
```

Step 2: Create Teacher Table

```
CREATE TABLE Teacher (
    Name VARCHAR(50),
    DeptNo INT,
    DateOfJoining DATE,
    DeptName VARCHAR(50),
    Location VARCHAR(50),
    Salary DECIMAL(10,2)
);
```

Explanation

- A table named **Teacher** is created.
- It stores teacher details like name, department, joining date, and salary.

Step 3: Insert Five Records

```
INSERT INTO Teacher VALUES
('Varun Chauhan', 101, '2018-06-15', 'Mathematics', 'Delhi', 50000),
('Jony Gautam', 102, '2019-07-20', 'Commerce', 'Mumbai', 45000),
('Akansha Yadav', 101, '2017-03-10', 'Mathematics', 'Delhi', 52000),
('Priya Singh', 103, '2020-01-05', 'Science', 'Chennai', 48000),
('Ankit Jain', 102, '2016-09-12', 'Commerce', 'Mumbai', 47000);
```

Explanation

- Five teacher records are inserted into the table.
- Each row contains department and salary details.

Step 4: Start Transaction

```
START TRANSACTION;
```

Explanation

- Begins a transaction.
- All changes after this can be either committed or rolled back.

Step 5: Increase Salary by 25% for Mathematics Department

```
UPDATE Teacher
SET Salary = Salary + (Salary * 0.15)
WHERE DeptName = 'Commerce';
```

Explanation

- Updates salary for teachers in **Mathematics** department.
- Adds **25% increment** to their salary.

Step 6: Perform ROLLBACK

```
ROLLBACK;
```

Explanation

- Cancels the previous update.
- Salaries return to original values (before 25% increment).

Step 7: Increase Salary by 15% for Commerce Department

```
UPDATE Teacher
SET Salary = Salary + (Salary * 0.15)
WHERE DeptName = 'Commerce';
```

Explanation

- Gives **15% salary increment** to Commerce department teachers.
- This change is still temporary until committed.

Step 8: Perform COMMIT

```
COMMIT;
```

Explanation

- Saves all changes permanently.
- Now the **15% increment cannot be undone**.

Step 9: Perform SELECT command to view the table teacher after commit.

- ```
  SELECT * FROM TEACHER;
  ```

Output:

Name varchar	DeptNo int	DateOfJoining date	DeptNam varchar	Location varchar	Salary decimal
Varun Chauhan	101	2018-06-15	Mathematics	Delhi	50000.00
Jony Gautam	102	2019-07-20	Commerce	Mumbai	51750.00
Akansha Yadav	101	2017-03-10	Mathematics	Delhi	52000.00
Priya Singh	103	2020-01-05	Science	Chennai	48000.00
Ankit Jain	102	2016-09-12	Commerce	Mumbai	54050.00
  ```